Accelerating metabolic engineering through multimodal phenotype prediction with TorchCell and the Cell Graph Transformer

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Abstract

Deep learning has advanced protein structure prediction, yet metabolic engineering lags because phenotype data are scattered across heterogeneous studies lacking a shared ontology. We present TorchCell, an open-source framework that annotates literature and public datasets with a shared ontology and ingests them into a Neo4j knowledge graph, then builds deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell combining genome sequence, gene networks, metabolism, and environment. We develop the Cell Graph Transformer (CGT), a virtual-cell architecture in which biological interaction graphs constrain multi-head attention and an equivariant perturbation operator feeds multitask decoders. In *Saccharomyces cerevisiae*, CGT predicts trigenic gene-gene interactions (Pearson $\boldsymbol{r} = \mathbf{0.454}$), outperforming DANGO, DCell, and Yeast9 flux balance analysis, and generalizes to knockout expression ($\boldsymbol{r} = \mathbf{0.543}$) and morphology ($\boldsymbol{r} = \mathbf{0.619}$). CGT recommends gene deletions for beta-carotene and betaxanthin production, pairing with autonomous biofoundries for AI-guided strain engineering.

Keywords: metabolic engineering, foundation models, gene interactions, virtual cell, knowledge graph, strain design

1 Introduction

[FILLER – replace with Intro prose.] Deep learning transformed protein structure and function prediction, yet systems biology and metabolic engineering lag because phenotype data are scattered across small, heterogeneous studies with no shared schema [1]. Prior approaches fall short: mechanistic flux balance analysis misses epistasis, and recent perturbation transformers do not consistently beat linear baselines [2]. Our own classical baselines predict knockout fitness but fail on gene interactions (Suppl. Fig. S1), motivating a deep, biologically grounded model. Here we present TorchCell and the Cell Graph Transformer (CGT), whose data model and architecture we outline in Suppl. Fig. S7.

2 Results

A shared ontology and knowledge graph unify metabolic-engineering data. *[FILLER – R1.]* TorchCell annotates literature and public datasets with a shared experiment ontology and ingests them into a Neo4j knowledge graph; dataloaders emit deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell (Fig. 1; data scale in Suppl. Fig. S2; ontology/KG spec in Suppl. Fig. S4).

The Cell Graph Transformer predicts trigenic gene-gene interactions at state of the art. *[FILLER – R2.]* The Cell Graph Transformer (CGT; architecture in Fig. 1c and Methods) constrains attention with biological interaction graphs and maps the reference cell through an equivariant perturbation operator. On trigenic gene-gene interactions (GGI), where classical ML fails (Suppl. Fig. S1), CGT reaches Pearson $r = 0.454$ (Spearman 0.421), outperforming Yeast9 FBA, DCell, and DANGO (Fig. 2).

One embedding, many phenotypes: expression, morphology, and fitness. *[FILLER – R3.]* The same embedding predicts knockout expression ($r = 0.543$), single-knockout morphology ($r = 0.619$), and fitness (Fig. 3; expression diagnostics in Suppl. Fig. S5).

Natural genetic variation versus model-designed perturbations. *[FILLER – R4.]* We compare phenotypes from natural strain genetic variation against deep-learning-designed gene perturbations across genome and environment (Fig. 4).

Extending to metabolism and strain design. *[FILLER – R5.]* Grounding the cell representation in metabolism, CGT recommends gene targets for production phenotypes and pairs with the UIUC iBioFoundry for iterative design (Fig. 5; inference at scale in Suppl. Fig. S6).

3 Discussion

[FILLER – Discussion.] A strain-aware, biologically grounded representation is the differentiator; limitations include single-organism scope, morphology-data maturity, and error-bar provenance.

4 Methods

Ontology and knowledge-graph construction

[FILLER.] Experiment ontology (genotype, phenotype, environment, experiment metadata); annotation of literature and public datasets; ingestion into the Neo4j knowledge graph; conversion, deduplication, and aggregation into HeteroData records (Suppl. Figs. S8, S9).

Datasets

[FILLER.] Source datasets and versions (e.g. Costanzo, Kuzmin trigenic; knockout expression; single-knockout morphology); inclusion/exclusion criteria; dataset sizes and label provenance.

Cell representation and the perturbation operator

[FILLER.] Shared wildtype reference cell; each strain as a perturbation operator over the reference; multimodal node/edge features (genome sequence, gene-interaction networks, metabolism, environment); DNA sequence selection and tokenization (Suppl. Figs. S10, S11).

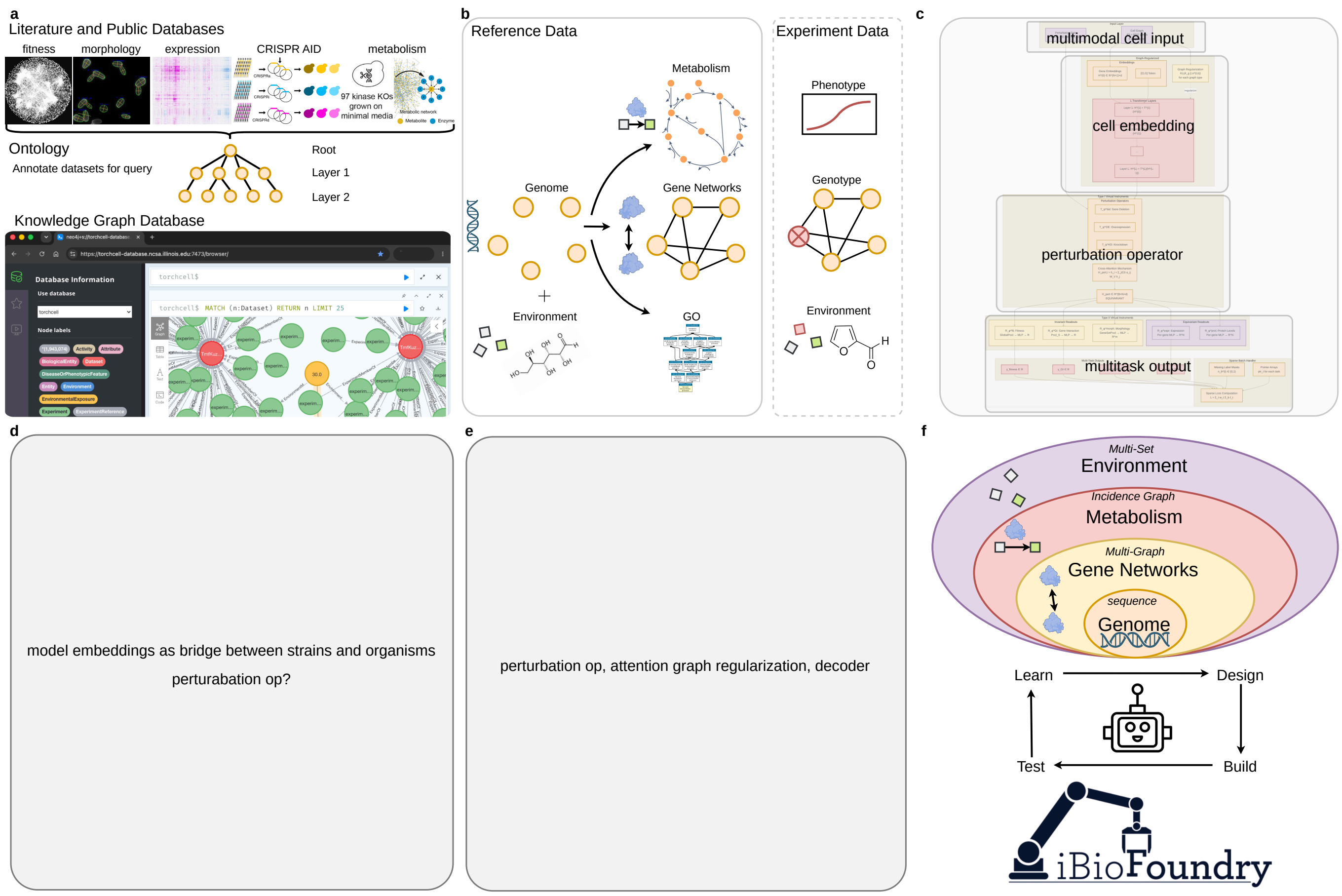

Fig. 1 TorchCell: a database and software framework for building deep-learning-ready metabolic-engineering datasets. (a) Literature and public databases are annotated with TorchCell's experiment ontology and ingested into a Neo4j knowledge graph (hosted at NCSA), queryable across datasets. (b) Genomic representation: reference data (genome, gene networks, metabolism, GO) and experiment data (phenotype, genotype, environment); experiments are compressed into perturbation operators over a shared reference state. (c) The Cell Graph Transformer (CGT) maps multimodal cell input to a cell embedding via a perturbation operator, decoded into multitask outputs. (d) Model embeddings bridge strains and organisms. (e) Perturbation operator, attention graph regularization, and decoder. (f) The multimodal cell representation feeds a Learn–Design–Build–Test loop with the NSF iBioFoundry.

Data splits and preprocessing

[FILLER.] Train/validation/test partitioning (and any gene- or interaction-disjoint splits); label normalization; handling of missing labels.

Cell Graph Transformer architecture

[FILLER.] Graph-constrained multi-head attention; the equivariant perturbation operator mapping reference embedding to perturbed state; multitask decoder heads (Fig. 1c). Layer counts, hidden dimensions, and parameter budget.

Graph-regularized attention and training objective

[FILLER.] Attention-prior alignment loss aligning attention to network priors; multitask loss weighting; full mathematical formulation.

Model training

[FILLER.] Optimizer, learning-rate schedule, batch size, early stopping; hardware (GPUs, distributed/DDP), wall-clock; software versions; random seeds.

[Fig. 2 placeholder – compose in draw.io]

Trigenic gene-gene interactions. (a) concept — why trigenic interactions are hard to predict; (b) predicted (CGT) vs. actual (experimental) interaction score; (c) state of the art vs. Yeast9 FBA / DCell / DANGO; (d) accuracy stratified by interaction magnitude $|\tau|$; (e) data and model scaling (r vs. #interactions and model size); (f) attention-graph-regularization ablation (does graph-constrained attention add value?).

Per plan: can split into two figures if crowded.

Fig. 2 State-of-the-art trigenic gene-gene interaction prediction. (a)–(f).

[Fig. 3 placeholder – compose in draw.io]

Expression, morphology, and fitness. (a) expression predicted vs. actual (top genes); (b) morphology predicted vs. actual; (c) fitness predicted vs. actual; (d) cross-phenotype summary / shared embedding; (e) trends across phenotype classes.

Fig. 3 One latent cell embedding generalizes across expression, morphology, and fitness. (a)–(e).

Baselines

[FILLER.] DANGO, DCell, and Yeast9 flux balance analysis; classical-ML baselines (Suppl. Fig. S1); matched training data and evaluation protocol.

Evaluation metrics

[FILLER.] Pearson correlation (r) and any secondary metrics; aggregation across folds; definition of error bars (see Statistics).

Gene–gene interaction prediction

[FILLER.] Trigenic interaction prediction setup; performance vs. interaction magnitude; ablations (Fig. 2).

Expression and morphology prediction

[FILLER.] Knockout transcriptome and single-knockout morphology tasks; shared latent embedding; cross-phenotype transfer (Fig. 3).

CGT-guided strain design

[FILLER.] Objective and candidate enumeration for beta-carotene and betaxanthin production; ranking and selection of recommended gene deletions (Fig. 5).

[Fig. 4 placeholder – compose in draw.io; from handwritten plan, confirm panels]

Natural variation vs. DL design. (a) natural strain genome variation and phenotype; (b) DL-predicted phenotype across the genome × environment landscape; (c) model-designed knockouts compared with natural variants.

Fig. 4 Natural genetic variation versus model-designed perturbations. (a)–(c).

[Fig. 5 placeholder – compose in draw.io; from handwritten plan, confirm panels]

Metabolism. (a) metabolic-network grounding / additional models; (b) benchmark; (c) kinase-knockout collection; (d) isobutanol; (e) tryptophan-pathway precursors; (f) beta-carotene / betaxanthin production with the iBioFoundry DBTL loop.

Fig. 5 Extending TorchCell to metabolism and metabolic-engineering strain design. (a)–(f).

Strain construction and cultivation

[FILLER.] Strains, plasmids, and genome edits; the TorchCell + UIUC iBioFoundry design–build–test–learn loop; cultivation and fermentation conditions.

Metabolite quantification

[FILLER.] Sampling, extraction, and analytical method (HPLC/LC–MS); calibration and titer quantification.

Inference at scale

[FILLER.] Large-scale inference pipeline and resources (Suppl. Fig. S6).

Statistics

[FILLER.] Statistical tests used and whether one- or two-sided; sample sizes (n); definition of center and dispersion; box-plot conventions (median, IQR hinges, whiskers $\leq$ 1.5×IQR); multiple-comparison handling; significance thresholds.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

[FILLER.] Datasets are available on Hugging Face (https://huggingface.co/...); the Neo4j

knowledge graph is hosted at NCSA (...); trained model weights are available at .... Source data are provided with this paper.

Code availability

[FILLER.] TorchCell and the Cell Graph Transformer are open source at https://github.com/Mjvolk3/torchcell (license: ...), with a release archived at Zenodo (https://doi.org/...).

Acknowledgments.

Declarations

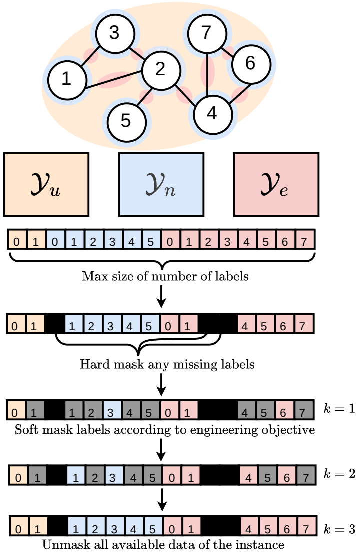

Fig. S7 TorchCell: ontology-unified knowledge graph and the perturbation-operator cell representation. Panels (a)–(d).

Supplementary information.

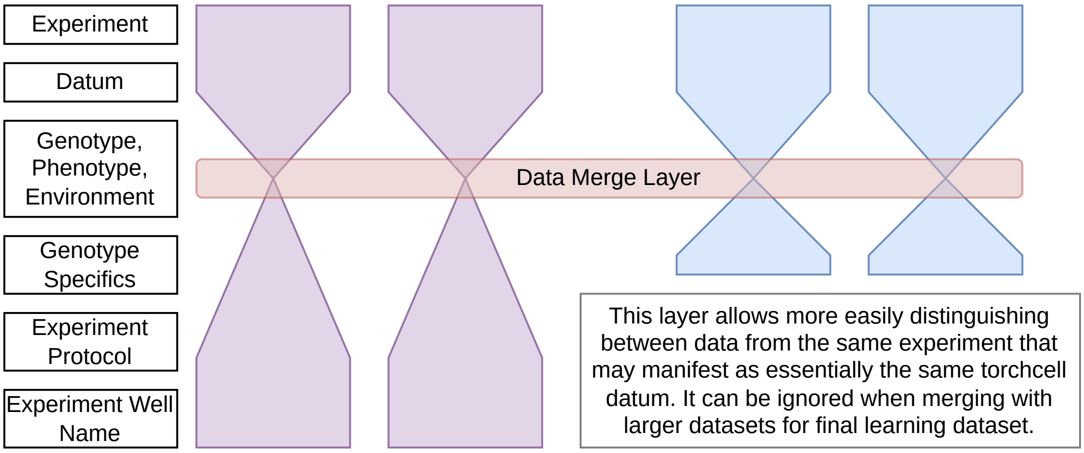

Fig. S8 Ontology data model (hourglass). Each experiment funnels genotype, phenotype, environment, and experiment-specific metadata through a shared Data Merge Layer that distinguishes records from the same experiment; the layer can be ignored when merging into larger datasets.

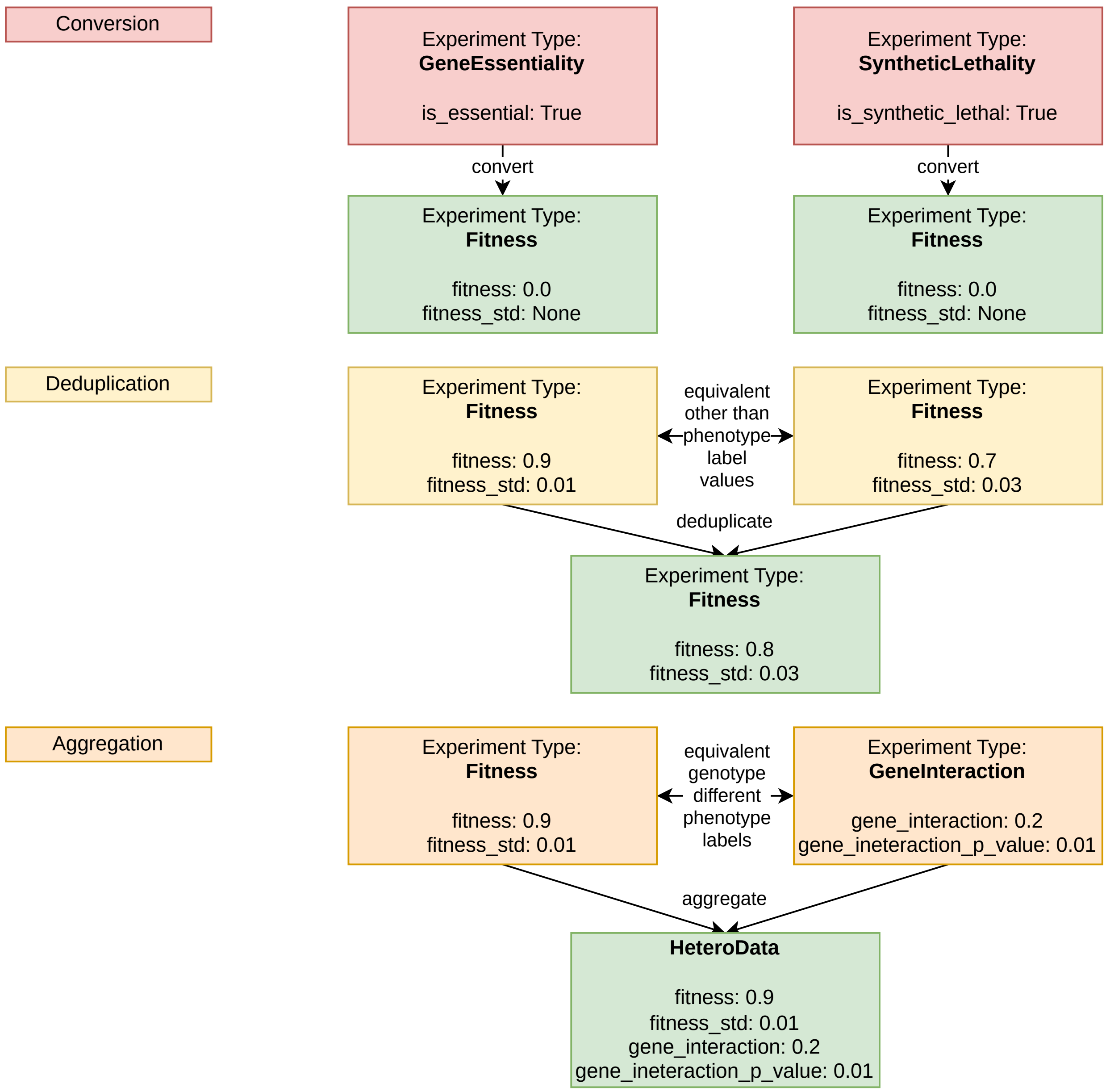

Fig. S9 Knowledge-graph normalization in three stages. Conversion maps phenotype experiment types (GeneEssentiality, SyntheticLethality) to a common Fitness type; deduplication collapses records equivalent other than their phenotype-label values; aggregation combines equivalent-genotype records with different phenotype labels (Fitness, GeneInteraction) into a single HeteroData instance.

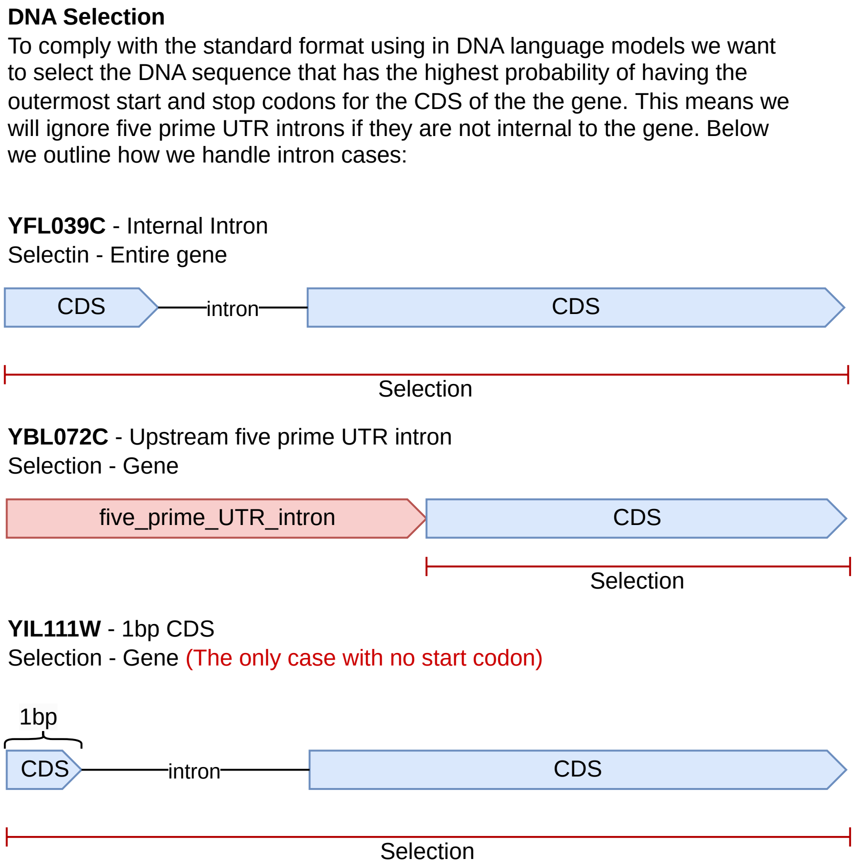

Fig. S10 DNA sequence selection for the sequence model. The coding sequence is taken from the outermost start to stop codon, ignoring five-prime UTR introns unless internal to the gene. Cases: internal intron (YFL039C), 5' UTR intron (YBL072C), and a 1 bp CDS lacking a start codon (YIL111W).

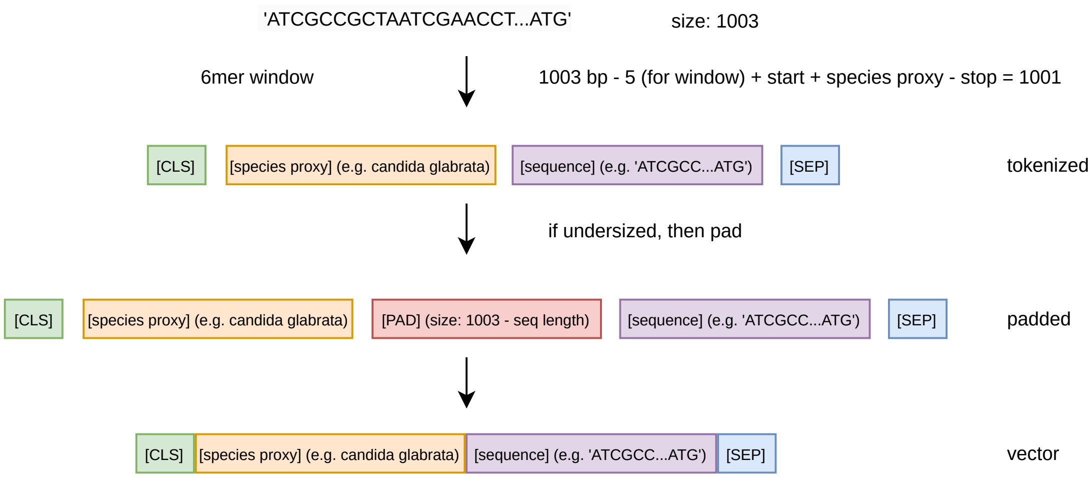

Fig. S11 DNA tokenization. A 1003 bp window is encoded as [CLS], species proxy, 6-mer sequence, [SEP]; undersized sequences are padded with [PAD] to a fixed length before vectorization.

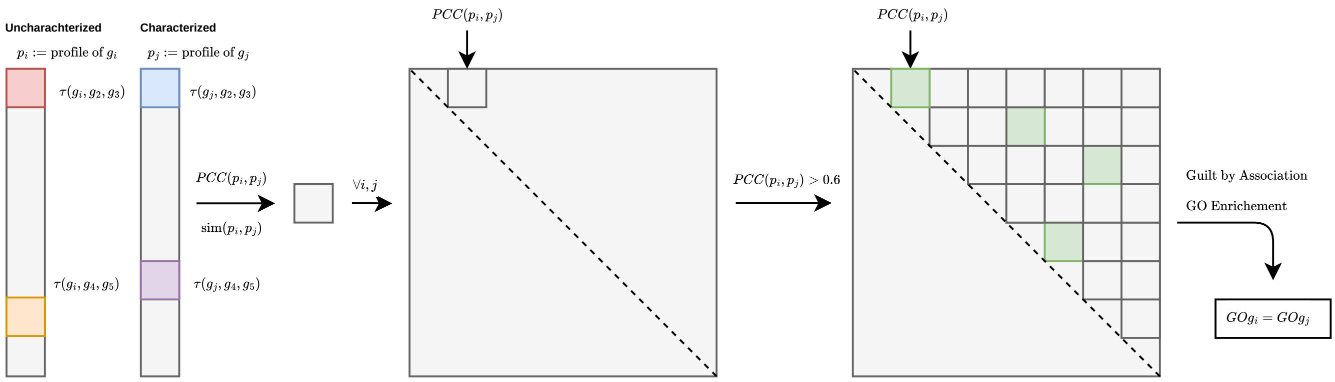

Fig. S12 Guilt-by-association annotation of uncharacterized genes. (a)–(e) Triple-interaction profiles p are correlated (Pearson, PCC) across all gene pairs; pairs with PCC > 0.6 are linked, and GO enrichment over the resulting network transfers annotations ($\mathrm{GO}_{g_i} = \mathrm{GO}_{g_j}$). (f) Total quantification of triple interactions across the remaining yeast genome (coverage of uncharacterized genes). (g) The resulting new annotations transferred to previously uncharacterized genes.

References

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